

UNIT I

Matrices

Eigen values and vectors of a real matrix characteristic equation. Properties of a Eigen values and vector. Caley Hamilton theorem, Diagonalization of matrices, Reductions of a quadratic form two canonical form by ortriogonal transformations nature of quadratic form

Caley Hamilton theorem:

Definition: Every square matrix a satisfy its own characteristics equation

Uses:

- We can calculate the +ve integral powers
- We can calculate the inverse of a square matrix A

Methods of finding characteristic equation:

Case 1: if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a square matrix of order 2, then the characteristics equations is $\lambda^2 - C_1\lambda + C_2 = 0$

- $C_1 =$ sum of the diagonal element
- $C_2 =$ determinant of matrix A.

Case 2: If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix of order 3, then the characteristic equation is $\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$

Where,

- $C_1 \rightarrow$ sum of the diagonal element
- $C_2 \rightarrow$ sum of the minor of the principle diagonal element
- $C_3 \rightarrow$ determinant of A

Note:

- $AA^{-1} = A^{-1}A = I$
- $A^{-1}I = A^{-1}$

1) Verify the CHT for the matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ and hence evaluate A^3 and A^{-1}

The characteristic equation of A is $\lambda^2 - C_1\lambda + C_2 = 0$

$$C_1 = 1 + 3 = 4$$

$$C_2 = 3 - 8 = -5$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\text{Sub } \lambda = A$$

$$A^2 - 4A - 5I = 0$$

$$A^2 = A \times A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 2+6 \\ 4+12 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 16 & 12 \end{bmatrix}$$

$$5I = 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\begin{aligned} A^2 - 4A - 5I &\Rightarrow \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 16 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Hence CHT is proved

To find A^3

Where is that: $A^2 - 4A - 5I = 0$

$$\Rightarrow \times A = A^3 - 4A^2 + 5IA = 0$$

Where

$$A^3 - 4A^2 + 5IA$$

$$4A^2 = 4 \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} = \begin{bmatrix} 36 & 32 \\ 64 & 68 \end{bmatrix}$$

$$5IA = 5 \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 20 & 15 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 36 & 32 \\ 64 & 68 \end{bmatrix} \begin{bmatrix} 5 & 10 \\ 20 & 15 \end{bmatrix} = \begin{bmatrix} 41 & 42 \\ 84 & 83 \end{bmatrix}$$

To find A^{-1}

$$\Rightarrow \times A^{-1} = A^2 A^{-1} - 4A A^{-1} - 5I A^{-1} = 0$$

$$A - 4I - 5A^{-1} = 0$$

$$\begin{aligned}
 5A^{-1} &= A - 4I \\
 &= \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \\
 5A^{-1} &= \begin{bmatrix} -3 & 2 \\ 4 & -1 \end{bmatrix} \\
 A^{-1} &= \frac{1}{5} \begin{bmatrix} -3 & 2 \\ 4 & -1 \end{bmatrix}
 \end{aligned}$$

2) Verify $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ in CHT and evaluate A^4 and A^{-1}

The characteristic equation $\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$

$$C_1 = 2 + 2 + 2 = 6$$

$$\begin{aligned}
 C_2 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\
 &= (4-1) + (4-1) + (4-1) \\
 &= 3 + 3 + 3 = 9
 \end{aligned}$$

$$C_3 = 2(3) + 1(-1) + 1(-1) = 4$$

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

Where

$$\lambda = A$$

$$A^3 - 6A^2 + 9A - 4I = 0$$

$$\begin{aligned}
 A^2 &= \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \\
 &= \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix} \\
 A^2 &= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}
 \end{aligned}$$

$$A^3 = A^2 \times A$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$6A^2 = 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 36 & -36 & 30 \\ -30 & 36 & -36 \\ 30 & -30 & 36 \end{bmatrix}$$

$$9A = 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 18 & -9 & 9 \\ 9 & 18 & 9 \\ 9 & -9 & 18 \end{bmatrix}$$

$$4I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{aligned} &= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -36 & 30 \\ -30 & 36 & -36 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ 9 & 18 & 9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Hence it is proved

$$A^3 - 6A^2 + 9A - 4I = 0$$

$$\times A \Rightarrow A^4 - 6A^3 + 9A^2 - 4AI = 0$$

$$A^4 - 6A^3 + 9A^2 - 4AI$$

$$6A^3 = 6 \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} = \begin{bmatrix} 132 & -126 & 126 \\ -126 & 132 & -126 \\ 126 & -126 & 132 \end{bmatrix}$$

$$9A^2 = 9 \begin{bmatrix} 6 & 5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 54 & -45 & 45 \\ -45 & 54 & -45 \\ 45 & -45 & 54 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & -4 & 4 \\ -4 & 8 & -4 \\ 4 & -4 & 8 \end{bmatrix}$$

$$\begin{aligned} A^4 &= \begin{bmatrix} 132 & -126 & 126 \\ -126 & 132 & -126 \\ 126 & -126 & 132 \end{bmatrix} - \begin{bmatrix} 54 & -45 & 45 \\ -45 & 54 & -45 \\ 45 & -45 & 54 \end{bmatrix} + \begin{bmatrix} 8 & -4 & 4 \\ -4 & 8 & -4 \\ 4 & -4 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 86 & -85 & 85 \\ -85 & 86 & -85 \\ 85 & -85 & 86 \end{bmatrix} \end{aligned}$$

$$\times A^{-1} \Rightarrow A^3 A^{-1} - 6A^{-1} A^2 + 9A A^{-1} - 4A^{-1} I = 0$$

$$A^2 - 6A + 9I - 4A^{-1} = 0$$

$$4A^{-1} = A^2 - 6A + 9I$$

$$4A^{-1} = \begin{bmatrix} 6 & 5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

3) Verify $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$ using CHT and find A^{-1}

$$\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$$

$$C_1 = 1 + 2 + 1 = 4$$

$$C_2 = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 7 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$$

$$= -4 - 6 - 10 = -20$$

$$C_3 = 1(2 - 6) - 3(4 - 3) + 7(8 - 2)$$

$$= -4 - 3 + 42$$

$$= 35$$

$$\lambda^3 - 4\lambda^2 - 20\lambda - 35 = 0$$

$$\text{Sub } \lambda = A \rightarrow A^3 - 4A^2 - 20A - 35I = 0$$

$$A \times A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+12+7 & 3+6+14 & 7+9+7 \\ 4+8+3 & 12+4+6 & 28+6+3 \\ 1+8+1 & 3+4+2 & 7+6+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix} \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 20 + 92 + 23 & 60 + 46 + 46 & 140 + 69 + 23 \\ 15 + 88 + 37 & 45 + 44 + 74 & 105 + 66 + 37 \\ 10 + 36 + 14 & 30 + 18 + 28 & 70 + 27 + 14 \end{bmatrix}$$

$$= \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix}$$

$$4A^2 = \begin{bmatrix} 80 & 92 & 92 \\ 60 & 88 & 148 \\ 40 & 36 & 56 \end{bmatrix}$$

$$20A = \begin{bmatrix} 20 & 60 & 140 \\ 80 & 40 & 60 \\ 20 & 40 & 20 \end{bmatrix}$$

$$35I = \begin{bmatrix} 35 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix} - \begin{bmatrix} 80 & 92 & 92 \\ 60 & 88 & 148 \\ 40 & 36 & 56 \end{bmatrix} - \begin{bmatrix} 20 & 60 & 140 \\ 80 & 40 & 60 \\ 20 & 40 & 20 \end{bmatrix} - \begin{bmatrix} 35 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 35 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\times A^{-1} \Rightarrow A^3 A^{-1} - 4A^2 A^{-1} - 20A A^{-1} - 35I A^{-1} = 0$$

$$A^2 - 4A - 20I - 35A^{-1} = 0$$

$$35A^{-1} = A^2 - 4A - 20I$$

$$A^{-1} = \frac{1}{35} \begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix}$$

4) Using CHT find A^{-1} if $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix}$

The characteristic equation $\lambda^3 - C_1 \lambda^2 + C_2 \lambda - C_3 = 0$

$$C_1 = 1 + 5 - 5 = 1$$

$$C_2 = \begin{bmatrix} 5 & -4 \\ 7 & -5 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 3 & -5 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

$$= (-25 + 28) + (-5 + 6) + (5 - 4)$$

$$= 3 + 1 + 1$$

$$= 5$$

$$C_3 = 1(-25 + 28) - 2(-10 + 12) - 2(14 - 15)$$

$$= 3 - 4 + 2 = 1$$

$$\lambda^3 - \lambda^2 + 5\lambda - I = 0$$

$$\text{Sub } \lambda = A \rightarrow A^3 - A^2 + 5A - I = 0$$

$$A^3 = \begin{bmatrix} -5 & -12 & 10 \\ -10 & -23 & 16 \\ -13 & -29 & 17 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{bmatrix}$$

$$5A = \begin{bmatrix} 5 & 10 & -10 \\ 1 & 25 & -20 \\ 15 & 35 & -25 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -5 & -12 & 10 \\ -10 & -23 & 16 \\ -13 & -29 & 17 \end{bmatrix} - \begin{bmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{bmatrix} + \begin{bmatrix} 5 & 10 & -10 \\ 1 & 25 & -20 \\ 15 & 35 & -25 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^3 - A^2 + 5A - I = 0$$

$$\Leftrightarrow \times A^{-1} = A^2 - A + 5AA^{-1} - IA^{-1} = 0$$

$$A^2 - A + 5I - A^{-1} = 0$$

$$A^{-1} = A^2 - A + 5I = 0$$

$$A^2 = \begin{bmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix}$$

$$5I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\begin{aligned}
 A^{-1} &= A^2 - A + 5I \\
 &= \begin{bmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} 3 & -4 & 2 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}
 \end{aligned}$$

Eigen values and Eigen vectors of a real matrix

Definition:

Eigen values/ proper values/ latent roots

Let $A = [a_{ij}]$ be a square matrix, the characteristic equations of A is $|A - \lambda I| = 0$, then the roots of the characteristic equations is called has the **Eigen** values

Eigen vectors/ latent vectors:

Let $A = [a_{ij}]$ be a square matrix, if there exist a non-zero vectors $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ for all $AX = \lambda X$, then the vector X is called as eigen vectors of A corresponding to the eigen values of λ

Working rule to find eigen values and vectors:

- To find the characteristic equation (i.e) $|A - \lambda I| = 0$
- To solve the CE we get characteristic roots. They are called eigen values
- To find eigen vectors solve $|A - \lambda I|X = 0$ for the different values of λ

Note: Type1 problem based on non-symmetric matrices with non-repeated eigen values

1) Find the eigen values and vectors of the matrix $A = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}$

Solution:

Let $A = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}$ CE of the matrix A is $|A - \lambda I| = 0$

$$\lambda^2 - S_1 \lambda + S_2 = 0$$

$$S_1 = 1 - 1 = 0$$

$$S_2 = -1 - 3 = -4$$

$$\lambda^2 - 4 = 0$$

$$\lambda^2 = 4$$

$$\lambda = -2$$

$$\lambda = 2, -2$$

Hence the eigen values are

$$\lambda_1 = 2$$

$$\lambda_2 = -2$$

To find eigen vectors, we have to solve $|A - \lambda I|X = 0$

$$\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1-\lambda & 1 \\ 3 & -1-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

When $\lambda = 2$

$$\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-X_1 + X_2 = 0 \rightarrow (1)$$

$$3X_1 - 3X_2 = 0 \rightarrow (2)$$

Take equation (1)

$$\frac{X_2}{1} = \frac{X_1}{1}$$

$$X_1 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Hence the corresponding eigen vectors

When $\lambda = -2$

$$\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$3X_1 + X_2 = 0 \rightarrow (3)$$

$$3X_1 + X_2 = 0 \rightarrow (4)$$

$$3X_1 = -X_2$$

$$\frac{x_1}{1} = \frac{-x_2}{3}$$

$$X_2 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Result:

- (i). The eigen values of a gn matrix A are 2, -2
- (ii). The eigen vectors corresponding to the eigen value $\lambda_1 = 2$ is $X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- (iii). The eigen vectors corresponding to the eigen values $\lambda_2 = -2$ is $X_2 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$

2) Find the eigen values of vectors of matrix $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$

Solution:

To find the characteristics equation

The CE is $|A - \lambda I|X = 0$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

$$S_1 = 1+2+3 = 6$$

$$S_2 = \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix}$$

$$= (6-2) + (3+2) + 2$$

$$= 4 + 5 + 2$$

$$S_2 = 11$$

$$S_3 = 1(6-2) - 0 - 1(2-4)$$

$$= 4+2$$

$$= 6$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\begin{array}{rrrr} 1 & -6 & 11 & -6 \\ 0 & 1 & -5 & 6 \\ \hline 1 & -5 & 6 & 0 \end{array}$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda_1 = 1 \quad \lambda_2 = 2 \quad \lambda_3 = 3$$

Eigen values are $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$

To find eigen vectors

$$(A - \lambda I) x = 0$$

$$\begin{pmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{pmatrix} x = 0$$

Table $\lambda_1 = 1$

$$\begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$x_3 = 0 \rightarrow (1)$$

$$x_1 + x_2 + x_3 = 0 \rightarrow (2)$$

$$2x_1 + 2x_2 + 2x_3 = 0 \rightarrow (3)$$

Table 1 of 2

	x_1	x_2	x_3
0	-1	0	0
1	1	1	1
1	-1	0	

$$\lambda_1 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

When $\lambda_2 = 2$

$$\begin{pmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-x_1 - x_3 = 0 \rightarrow x_1 = -x_3$$

$$X_1 + x_3 = 0 \rightarrow (5)$$

$$2x_1 + 2x_2 + 2x_3 = 0 \rightarrow (6)$$

Equation (4) of (5) are same so we use (4) of (6)

$$X_1 + 0x_2 + x_3$$

$$2x_1 + 2x_2 + x_3$$

$$x_1 \quad x_2 \quad x_3$$

$$0 \quad 1 \quad 1 \quad 0$$

$$2 \quad 1 \quad 2 \quad 2$$

$$-2 \quad 1 \quad 2$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$$

$$\lambda_3 = 3$$

$$\begin{pmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-2x_1 + 0x_2 - x_3 \rightarrow (7)$$

$$x_1 - x_2 + x_3 \rightarrow (8)$$

$$2x_1 + 2x_2 + 0x_3 \rightarrow (9)$$

$$x_1 \quad x_2 \quad x_3$$

$$0 \quad -1 \quad -2 \quad 0$$

$$-1 \quad 1 \quad 1 \quad -1$$

$$-1 \quad 1 \quad 2$$

$$\lambda_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

$$\lambda_1 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

$$\lambda_2 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

$$\lambda_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Type 2:

Problems based on non-symmetric matrices with repeated eigen values.

2) Find eigen values of vectors for the matrix $A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$

To find the CE

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

$$S_1 = -2 + 1 + 0 = -1$$

$$S_2 = \begin{pmatrix} 1 & -6 \\ -2 & 0 \end{pmatrix} + \begin{pmatrix} -2 & -3 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} -2 & 2 \\ 2 & 1 \end{pmatrix}$$

$$= -12 - 3 - 6$$

$$= -21$$

$$S_3 = -2(-12) - 2(-6) - 3(-4+1)$$

$$= 24 + 12 + 9$$

$$= 45$$

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

-3	1	1	-21	-45
	0	-3	6	46

$$1 \quad -2 \quad -15 \quad 0$$

$$\lambda^2 - 2\lambda - 15 = 0$$

Eigen values are:

$$\lambda_1 = -3, \lambda_2 = -3, \lambda_3 = 5$$

$$\lambda_3 = 5$$

$$\begin{pmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -\lambda & -2 & -\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-7x_1 + 2x_2 - 3x_3 \rightarrow (1)$$

$$2x_1 - 4x_2 - 6x_3 \rightarrow (2)$$

$$-1x_1 - 2x_2 - 5x_3 \rightarrow (3)$$

$$\begin{array}{rrrr} & x_1 & x_2 & x_3 \\ -4 & -6 & 2 & -4 \\ -2 & -5 & -1 & -2 \\ & 8 & 16 & -8 \\ \div 8 \rightarrow & 1 & 2 & -1 \end{array}$$

$$\lambda = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

When $\lambda_2 = -3$

$$\begin{pmatrix} -1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$2x_1 + 4x_1 - 6x_3 \Rightarrow x_1 + 2x_2 - x_3 = 0$$

$$-x_1 - 2x_2 - 3x_3 \Rightarrow 0$$

$$x_1 = 0 \Rightarrow 2x_2 = 3x_3$$

$$\frac{x_2}{3} = \frac{x_3}{2}$$

$$\lambda_2 = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}$$

Put $x_2 = 0$

$$x_1 = 3x_3$$

$$\frac{x_1}{3} = \frac{x_3}{1}$$

$$\lambda_3 = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda_1 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}$$

$$\lambda_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

Find the eigen values of vectors for the matrix $A = \begin{pmatrix} 6 & -6 & 5 \\ 14 & -13 & 10 \\ 7 & -6 & 4 \end{pmatrix}$

To find characteristic equation

$$|A - I \lambda| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

$$S_1 = 6 - 13 + 4 = -3$$

$$S_2 = \begin{pmatrix} 13 & 10 \\ -6 & 4 \end{pmatrix} + \begin{pmatrix} 6 & 5 \\ 7 & 4 \end{pmatrix} + \begin{pmatrix} 6 & -6 \\ 14 & -13 \end{pmatrix}$$

$$= (-52 + 60) + (24 - 35) + (-78 + 84)$$

$$= 8 + (-11) + 6$$

$$= 3$$

$$S_3 = 6(-52 + 60) + 6(56 - 70) + 5(84 - 91)$$

$$= 6(8) - 6(14) - 5(7)$$

$$= 48 - 84 - 35$$

$$= -71$$

$$\lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$$

$$\begin{array}{ccccc} -1 & 1 & 3 & 3 & 1 \\ & 0 & -1 & -2 & -1 \\ & -1 & 2 & 1 & 0 \end{array}$$

$$\lambda = -1$$

$$\begin{pmatrix} 7 & -6 & 5 \\ 14 & -12 & 10 \\ 7 & -6 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$7x_1 - 6x_2 + 5x_3 = 0$$

$$7x_1 - 6x_2 + 5x_3 = 0$$

$$7x_1 - 6x_2 + 5x_3 = 0$$

$$X_1 = 0$$

$$X_2 = 5$$

$$X_3 = 6$$

$$X_1 = \begin{pmatrix} 0 \\ 5 \\ 6 \end{pmatrix}$$

$$X_2 = 0 \quad 7x_1 = -5x_3$$

$$X_2 = \begin{pmatrix} -5 \\ 0 \\ 7 \end{pmatrix}$$

$$X_3 = 0 \quad 7x_1 = 6x_2$$

$$x_3 = \begin{pmatrix} 6 \\ 7 \\ 0 \end{pmatrix}$$

Find the eigen values of vectors of the matrix $A = \begin{pmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{pmatrix}$

To find CE

$$|A - I\lambda| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

$$S_1 = 7 + 6 + 5$$

$$= 18$$

$$S_2 = \begin{pmatrix} 6 & -2 \\ -2 & 5 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 5 \end{pmatrix} + \begin{pmatrix} 7 & -2 \\ -2 & 6 \end{pmatrix}$$

$$= 30 - 4 + 35 + (42 - 4)$$

$$= 26 + 35 + 38$$

$$= 99$$

$$S_3 = 7(30 - 4) + 2(-10) + 0$$

$$= 182 - 20$$

$$S_3 = 162$$

$$\lambda^3 - 18\lambda^2 + 99\lambda - 162 = 0$$

$$\lambda_1 = 9$$

$$\lambda_2 = 3$$

$$\lambda_3 = 6$$

$$\lambda = 9$$

$$\lambda^3 - 18\lambda^2 + 99\lambda - 162 = 0$$

$$\begin{pmatrix} -2 & -2 & 0 \\ -2 & -3 & -2 \\ 0 & -2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-2X_1 - 2X_2 + 0X_3 \rightarrow (1)$$

$$-2X_1 - 3X_2 - 2X_3 \rightarrow (2)$$

$$0X_1 - 2X_2 - 4X_3 \rightarrow (3)$$

Taking (1) of (2) we get

	X_1	X_2	X_3
-2	0	-2	-2
-3	-2	-2	-3
	4	-4	2
$\div 2 \Rightarrow 2$		-2	1

$$\lambda_1 = \begin{pmatrix} 2 \\ -2 \\ . \end{pmatrix}$$

$$\lambda_2 = 3$$

$$\begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$4X_1 - 2X_2 + 0X_3$$

$$-2X_1 + 3X_2 - 2X_3$$

$$0X_1 - 2X_2 + 2X_3$$

$$\begin{array}{cccc} & x_1 & x_2 & x_3 \\ -2 & 0 & 4 & -2 \\ 3 & -2 & -2 & 3 \\ & 4 & 8 & 8 \\ \div 4 \Rightarrow & 1 & 2 & 1 \end{array}$$

$$X_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$\lambda = 6 \begin{pmatrix} 1 & -2 & 0 \\ -2 & 0 & -2 \\ 0 & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$X_1 - 2X_2 + 0X_3$$

$$-2X_1 + 0X_2 - 2X_3$$

$$0X_1 - 2X_2 - X_3$$

$$\begin{array}{cccc} & X_1 & X_2 & X_3 \\ -2 & 0 & 1 & -2 \\ 0 & -2 & -2 & 0 \\ & 4 & 2 & 4 \\ \div 2 \Rightarrow & 2 & 1 & -2 \end{array}$$

$$X_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Properties of eigen values of vectors

- 1) A square matrix A and its transpose have same eigen values.
- 2) The sum of the eigen values of a matrix A is equal to the sum of the element in its diagonal.

- 3) If all the eigen value that is $\lambda_1, \lambda_2, \lambda_3 \dots \dots \dots \lambda_n$ of will be linearly independent.
- 4) If two or more eigen values of a matrix A are to the eigen vectors which may be either linearly dependent or linearly independent.
- 5) The product of the eigen value
- 6) If $\lambda_1, \lambda_2, \lambda_3 \dots \dots \dots \lambda_n$ are eigen values of a matrix A than A^{-1} has the eigen values as $1/\lambda_1, 1/\lambda_2, \dots \dots \dots 1/\lambda_n$
- 7) If $\lambda_1, \lambda_2, \dots \dots \dots \lambda_n$ are the eigen values of the matrix A, then A^n has the eigen values as $\lambda_1^n, \lambda_2^n \dots \dots \dots \lambda_n^n$
- 8) The eigen vectors corresponding to the distinct eigen values of a real symmetric matrix (or) orthogonal.

- 1) If the eigen values of $A = \begin{pmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{pmatrix}$ are 2, 2, 3 find the eigen value of A^{-1} eigen values of $A^{-1} = 1/2, 1/2, 1/3$

- 2) Find the eigen values of A^5 when $A = \begin{pmatrix} 3 & 0 & 0 \\ 5 & 4 & 0 \\ 3 & 6 & 1 \end{pmatrix}$ the values are 1,3,4

The value of $A^5 = 1^5, 3^5, 4^5,$

$$= 1, 243, 1024$$

- 3) Find the sum of product of all the eigen values of matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$

$$S1 = 1+2+3 = 6 \Rightarrow \text{Sum}$$

$$S2 = 1(6-4) - 1(3-2) + 1(2-2) \Rightarrow \text{product}$$

$$= 2-1$$

$$= 1$$

- 4) If two of the eigen values of matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ are 2 or 8 find the 3rd eigen values

$$\lambda_1 + \lambda_2 + \lambda_3 = 12$$

$$2 + 8 + \lambda_3 = 12$$

$$\lambda_3 = 12 - 10$$

$$\lambda_3 = 2$$

From property

Sum of eigen values = sum of diagonal element diagonalisation of a matrix

Diagonalisation of a matrix

Prop 1:

Diagonalisation of a matrix A is the processes of reduction of matrix A to a diagonal form. The process of reduction of matrix to a diagonal form is as follows.

- i) A square matrix A of order n with n linearly independent eigen vectors can be diagonalised by a similarity transformation (i.e) $D = B^{-1}AB$. Where B is the modal matrix whose column are the eigenvectors of A.

Note (i):

If D is a diagonal matrix then its eigen values are same as the diagonal elements.

- ii) Calculation of a power matrix if the transformation is A^n then

$$A^n = BD^n B^{-1}$$

Problem:

1). Diagonalise the matrix $A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$ of find A^4

Solution:

The CE of A is $(A - \lambda I) = 0$

$$(i.e) \lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

$$S_1 = 11$$

$$S_2 = 14 + 8 + 14$$

$$= 36$$

$$S_3 = 4 + (-2) + (-4)$$

$$= 36$$

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

$$\lambda_1 = 6$$

$$\lambda_2 = 3$$

$$\lambda_3 = 2$$

To find eigen vectors

$$(A - \lambda I)X = 0$$

$$\begin{pmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

When $\lambda = 6$

$$\begin{pmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-3X_1 - X_2 + X_3 = 0$$

$$-X_1 - X_2 - X_3 = 0$$

$$X_1 - X_2 - 3X_3 = 0$$

Take (1) & (2)

	x1	x2	x3
-1	1	-3	-1
-1	-1	-1	-1
	2	-4	2

$$X_1 = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} \text{ or } \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$$\text{When } \lambda=3 \quad \begin{pmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x1 \\ x2 \\ x3 \end{pmatrix} = 0$$

$$-X_2 + X_3 = 0$$

$$-X_1 + 2X_2 - X_3 = 0$$

$$X_1 - X_2 = 0$$

Take (4) of (5)

	X1	X2	X3
-1	1	0	-1
2	-1	-1	2
	-1	-1	-1

$$X_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

When $\lambda = 2$

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$X_1 - X_2 + X_3 = 0$$

$$-X_1 + 3X_2 - X_3 = 0$$

$$X_1 - X_2 + X_3 = 0$$

Take (7) & (8)

	X_1	X_2	X_3
-1	1	1	-1
3	-1	-1	3
	-2	0	2

$$X_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

When $\lambda = 6$

$$X_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

When $\lambda = 3$

$$X_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

When $\lambda = 2$

$$X_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Hence the model matrix B is $\begin{pmatrix} 1 & 1 & -1 \\ -2 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

$$D = B^{-1} AB$$

$$B^{-1} = \frac{\text{adj } B}{|B|}$$

adj B = Co – factor of B^T

$$\text{Co-factor of } B = \begin{pmatrix} 1 & (-2) & -3 \\ -2 & 2 & 0 \\ 1 & -(-2) & 3 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 1 & -2 & 1 \\ 2 & 2 & 2 \\ -3 & 0 & 3 \end{pmatrix}$$

$$|B| = 1(1) - 1(-2) - (-2-1)$$

$$= 1 + 2 + 3$$

$$= 6$$

$$B^{-1} = 1/6 \begin{pmatrix} 1 & -2 & 1 \\ 2 & 2 & 2 \\ -3 & 0 & 3 \end{pmatrix}$$

$$D = B^{-1} AB$$

$$= 1/6 \begin{pmatrix} 1 & -2 & 1 \\ 2 & 2 & 2 \\ -3 & 0 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$D = \frac{1}{6} \begin{pmatrix} 36 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 12 \end{pmatrix}$$

$$D = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

To find A^4

$$A^n = BD^nB^{-1}$$

$$A^4 = BD^4B^{-1}$$

$$D^4 = \begin{pmatrix} 6^4 & 0 & 0 \\ 0 & 3^4 & 0 \\ 0 & 0 & 2^4 \end{pmatrix}$$

$$D^4 = \begin{pmatrix} 1296 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 16 \end{pmatrix}$$

$$= \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1296 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 16 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 2 & 2 & 2 \\ -3 & 0 & 3 \end{pmatrix}$$

$$A^4 = \frac{1}{6} \begin{pmatrix} 1506 & -2430 & 1410 \\ -2430 & 5346 & -2430 \\ 1410 & -2430 & 1506 \end{pmatrix}$$

$$= \begin{pmatrix} 351 & -405 & 235 \\ -405 & 841 & -405 \\ 235 & -405 & 351 \end{pmatrix}$$

Diagonalize by similarity transform mission $A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{pmatrix}$

Solution:

The CE of A is $|A - \lambda I| = 0$

$$(i.e) \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$S_1 = 0$$

$$S_2 = -5 + 6 = 1$$

$$S_3 = -10 - 2 = -12$$

Quadratic Form:

The homogeneous polynomial of the 2nd degree in any number of variables is called a quadratic form

Example

$$X_1^2 + 2X_2^2 - 3X_3^2 + 5X_1X_2 - 6X_1X_3 + 4X_2X_3$$

Is a quadratic form in 3 variables

- The rank of a matrix A is called the rank of the quadratic form.
- If $r < n$ (or) $|A| = 0$ then the quadratic form is called singular.

Linear transformation of a quadratic form:

Let $Q = X^TAX$ be a quadratic form in the n variables (i.e.), $X_1, X_2, \dots, X_n$ consider the transformation $X = PY$ that transforms the variable set $X = (X_1, X_2, \dots, X_n)^T$ to a new variable set $Y = (Y_1, Y_2, \dots, Y_n)^T$ where T is a Non-singular matrix

We can easily verify that the transformation $X = PY$ expresses each of the variable which is called as non-singular linear transformation.

By this transformation (i.e.), $Q = X^TAX$ is transformed to $B = P^TAP$

Therefore B is symmetric matrix (i.e.), $B^T = B$

Then Y^TBY is the linear transformation of the quadratic form X^TAX under the transformation

$$B = P^TAP$$

Canonical Form:

It is of the quadratic equations. In the linear transformation $X = PY$, P is chosen such that

$B = P^TAP$ is a diagonal matrix of the form $\begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$ then the quadratic form Q gets

reduced as $Q = Y^TBY$ that is $Q = \lambda_1 \lambda_1^2 + \lambda_2 \lambda_2^2 + \dots + \lambda_n \lambda_n^2$ this form of Q is called sum of the square form of Q as canonical form of Q

Orthogonal reduction of a quadratic form to the canonical form:

In the transformation $X = PY$, P is the orthogonal matrix and if $X = PY$ transforms the quadratic form Q to the canonical form then Q is said to be reduced to the canonical form by an

orthogonal transformation or principle axis form. If N is an orthogonal matrix, the transformation is called an orthogonal linear transformation.

Let X^TAX be a quadratic form, let N be a normalized model matrix of A the matrix $D = N^TAN$ is a diagonal matrix with eigen values of A as diagonal elements

Working rule:

- Write the matrix of the given quadratic form
- To find the CE
- To solve the CE
- To find the eigen vectors orthogonal to each other
- Form normalized matrix N
- Find N^T
- Find $D = N^TAN$
- Find canonical form (i.e) ..., $(Y_1, Y_2, Y_3) (D) \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$

Nature of a quadratic form:

Let $Q = X^TAX$ be a quadratic form in N variables (i.e) ..., $X_1, X_2, \dots, X_n$.

Index of the quadratic form:

Definition:

The number of $+ve$ square terms in the canonical form is called the index of the quadratic form

The quadratic form in n variables is said to be

- $+ve$ definite if all the eigen values of A are $+ve$ nos
- $-ve$ definite (i.e) ..., if all the eigen values of A are $-ve$ nos

- +^{ve} semi definite (i.e) if all the eigen values of $A \geq 0$ and at least 1 eigen is zero.
- -^{ve} semi definite (i.e) if all the eigen values of $A \leq 0$ and atleast 1 eigen values is zero.
- Indefinite (i.e) matrix A has both the and -^{ve} values

Orthogonal matrix:

A square matrix A with real elements is said to be orthogonal if $AA^1 = A^1A = I$ since $AA^{-1} = A^{-1}A = I$ if follows $A^1 = A$

1). Determine the nature of the following quadratic form (i.e) $X_1^2 + 2X_2^2$

Solution:

The matrix of the quadratic form

Formula:

$$\begin{pmatrix} \text{Co-efficient of } X_1^2 & \frac{1}{2} \text{co-eff } X_1X_2 & \frac{1}{2} \text{co-eff } X_1X_3 \\ \frac{1}{2} \text{co-eff } X_2X_1 & \text{co-eff of } X_2^2 & \frac{1}{2} \text{co-eff } X_2X_3 \\ \frac{1}{2} \text{co-eff } X_3X_1 & \frac{1}{2} \text{co-eff } X_3X_2 & \text{co-eff of } X_3^2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{(i.e) Matrix is } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$D_1 = |a_{11}| = 1 \text{ (+ve)}$$

$$D_2 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = 2 \text{ (+ve)}$$

$$D_3 = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = 0 \quad (+^{\text{ve}} \text{ semi definite})$$

$$D_1 = 1, D_2 = 2, D_3 = 0.$$

Therefore Quadratic equation is $+^{\text{ve}}$ semi definite

2). Determine the nature of the following quadratic form

$$X^2 + 2Y^2 + 3Z^2 + 2XY + 2YZ - 2XZ$$

$$\text{Matrix is } \begin{pmatrix} 1 & 2/2 & -2/2 \\ 2/2 & 2 & 2 \\ -2/2 & 2/2 & 3 \end{pmatrix}$$

$$D_1 = |a_{11}| = 1$$

$$D_2 = 2 - 1 = 1$$

$$D_3 = 1(6-1) - 1(3+1) - 1(1+2)$$

$$= 5 - 4 - 3$$

$$D = -2$$

Therefore quadratic equation is indefinite

3). Discuss the nature $2X^2 + 3Y^2 + 2Z^2 + 2XY$

$$= \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$D_1 = |a_{11}| = 2$$

$$D_2 = 6 - 1 = 1$$

$$D_3 = 2(6) - 1(2) + (0)$$

$$= 12 - 2$$

$$D = 10$$

11m

Problems based on orthogonal reduction of a quadratic form to its canonical form

1). Reduce the quadratic form $Q = 6X^2 + 3Y^2 + 3Z^2 - 4XY - 2YZ + 4ZX$ into canonical form by an orthogonal transformation and discuss its nature

Solution:

gn Quadratic form is $6X^2 + 3Y^2 + 3Z^2 - 4XY - 2YZ + 4ZX$

The matrix of the quadratic form is $= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

To find the CE of a matrix (i.e), $\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$

$$C_1 = 12$$

$$C_2 = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 6 & -2 \\ -2 & 2 \end{pmatrix}$$

$$= (9-1) + (18-4) + (18-4)$$

$$= 8 + 14 + 14$$

$$= 36$$

$$C_3 = 6(9-1) + 2(-6+2) + 2(2-6)$$

$$= 48 - 8 - 8$$

$$= 32$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\lambda_1 = 8$$

$$\lambda_2 = 2$$

$$\lambda_3 = 2$$

To eigen vectors:

$$\text{When } \lambda = 8 \begin{pmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{pmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-2X_1 - 2X_2 + 2X_3 = 0$$

$$-2X_1 - 5X_2 - 1X_3 = 0$$

$$2X_1 - X_2 - 5X_3 = 0$$

$$\begin{array}{ccc} X_1 & X_2 & X_3 \end{array}$$

$$\begin{array}{cccc} -2 & +2 & -2 & -2 \end{array}$$

$$\begin{array}{cccc} -5 & -1 & -2 & -5 \end{array}$$

$$\begin{array}{cccc} 12 & -6 & 6 \end{array}$$

$$X_1 = \begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

When $\lambda = 2$

$$\begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$4X_1 - 2X_2 + 2X_3 = 0 \rightarrow 2X_1 - X_2 + X_3$$

$$2X_1 + X_2 - X_3 = 0$$

$$2X_1 - X_2 + X_3 = 0$$

X_1	X_2	X_3	
-2	+2	4	-2
1	-1	-2	1
0			

Sub $X_1 = 0$

$$-X_2 + X_3 = 0$$

$$\frac{X_2}{1} = \frac{X_3}{1}$$

$$X_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

To find 3rd eigen vector orthogonal to X_1 and X_2 , since A is symmetric, let $X_3 = L, m, n$

$$X_3 = \begin{pmatrix} l \\ m \\ n \end{pmatrix}$$

By (1) and (2)

$$2l - m + n = 0$$

$$m + n = 0$$

l	m	n	
-1	1	2	1
1	1	0	1

$$\begin{array}{ccc} 2 & -2 & 2 \\ \hline \end{array}$$

$$X_3 = \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

To form the normalized matrix

$$N = \begin{pmatrix} X_1/l \\ X_2/l \\ X_3/l \end{pmatrix} \text{ (i.e.) } l = \sqrt{X_1^2 + X_2^2 + X_3^2}$$

$$X_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

$$l = \sqrt{2^2 + (-1)^2 + 1^2}$$

$$= \sqrt{4 + 1 + 1}$$

$$= \sqrt{6}$$

$$X_1 = \begin{pmatrix} \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$$

$$X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$l = \sqrt{0 + 1 + 1}$$

$$= \sqrt{2}$$

$$X_2 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$X_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

$$l = \sqrt{(-1)^2 + (-1)^2 + 1^2}$$

$$l = \sqrt{3}$$

$$X_3 = \begin{pmatrix} \frac{-1}{\sqrt{3}} \\ \frac{-1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$N = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{pmatrix}$$

$$N^T = \begin{pmatrix} 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$$D = N^T A N$$

$$= \begin{pmatrix} 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

To find its canonical form that it (Y_1, Y_2, Y_3) $D = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$

(i.e)..,

$$(Y_1, Y_2, Y_3) \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$$

$$(8Y_1, 2Y_2, 2Y_3) \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$$

$$8Y_1^2 + 2Y_2^2 + 2Y_3^2$$

Canonical form is
 $8Y_1^2 + 2Y_2^2 + 2Y_3^2$

To find its nature:

$$Q = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

$$D_1 = |6| = 6$$

$$D_2 = \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} = 18 - 4 = 14$$

$$D_3 = 6(9-1) + 2(-6+2) + 2(2-6) \\ = 32$$

Nature is +ve definite

2). Reduce the quadratic form $2XY + 2YZ + 2ZX$ into canonical form and discuss its nature

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}$$

To find CE of matrix

$$\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$$

$$C_1 = 0$$

$$C_2 = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$= -4 - 4 - 4$$

$$= -12$$

$$C_3 = 0 - 2(-4) + 2(4)$$

$$= 8 + 8$$

$$= 16$$

$$\lambda^3 - 12\lambda^2 - 32 = 0$$

$$\lambda_1 = 4$$

$$\lambda_2 = -2$$

$$\lambda_3 = -2$$

To eigen vectors: When $\lambda = 4$

$$\begin{pmatrix} 0 - \lambda & 2 & 2 \\ 2 & 0 - \lambda & 2 \\ 2 & 2 & 0 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{pmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-4X_1 + 2X_2 + 2X_3 = 0 \rightarrow -2X_1 + X_2 + X_3$$

$$2X_1 - 4X_2 + 2X_3 = 0 \rightarrow X_1 - 2X_2 + X_3$$

$$2X_1 + 2X_2 - 4X_3 = 0 \rightarrow X_1 + X_2 - 2X_3$$

X_1	X_2	X_3	
1	1	-2	1
-2	1	1	-2
3	3	3	

$$X_1 = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

When $\lambda = -2$

$$\begin{pmatrix} -2 & 2 & 2 \\ 2 & -2 & 2 \\ 2 & 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-2X_1 + 2X_2 + 2X_3 = 0$$

$$2X_1 - 2X_2 + 2X_3 = 0$$

$$2X_1 + 2X_2 - 2X_3 = 0$$

Sub $X_1 = 0$

$$\frac{X_2}{1} = -\frac{X_3}{1}$$

$$X_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$X_3 = \begin{pmatrix} l \\ m \\ n \end{pmatrix}$$

By (1) and (2)

$$l + m + n$$

$$m - n = 0$$

$$l_1 = 0$$

X_1	X_2	X_3	
1	1	1	1
1	-1	0	1
-2	1	1	

$$X_3 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

$$N = \begin{pmatrix} 1/\sqrt{3} & 0 & -2/\sqrt{3} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{pmatrix}$$

$$N^T = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -2/\sqrt{6} & 1/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}$$

$$D = N^T A N$$

The diagonal matrix is $D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$$(Y_1, Y_2, Y_3) \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$$

$$= 2Y_1^2 - Y_2^2 - Y_3^2$$

Nature:

$$Q = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$D_1 = 0$$

$$D_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$$

$$D_3 = -2$$

$\Rightarrow$ Indefinite

3). Reduce the quadratic form $Q = 10X_1^2 + 2X_2^2 + 5X_3^2 + 6X_2X_3 - 10X_3X_1 - 4X_1X_2$ to a canonical form by orthogonal reduction and discuss its nature.

$$\begin{pmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{pmatrix}$$

$$\lambda^3 - C_1\lambda^2 + C_2\lambda - C_3 = 0$$

$$C_1 = 17$$

$$C_2 = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix} + \begin{pmatrix} 10 & -5 \\ -5 & 5 \end{pmatrix} + \begin{pmatrix} 10 & -2 \\ -2 & 2 \end{pmatrix}$$

$$= (10-9) + (50-25) + (20-4)$$

$$= 1 + 25 + 16$$

$$C_2 = 42$$

$$C_3 = 10(10-9) + 2(-10+15) - 5(-6+10)$$

$$= 10 + 10 - 20$$

$$= 0$$

$$\lambda^3 - 17\lambda^2 + 42\lambda = 0$$

$$\lambda (\lambda^2 - 17\lambda + 42) = 0$$

$$\lambda = 0, 14, 3$$

When $\lambda = 0$,

$$\begin{pmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{pmatrix} = 0$$

$$\begin{pmatrix} 10 - \lambda & -2 & -5 \\ -2 & 2 - \lambda & 3 \\ -5 & 3 & 5 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$10X_1 - 2X_2 - 5X_3 = 0$$

$$-2X_1 + 2X_2 + 3X_3 = 0$$

$$-5X_1 + 3X_2 + 5X_3 = 0$$

X_1	X_2	X_3	
-2	-5	10	-2
2	3	-2	2
4	-20	16	

$$X_1 = \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}$$

When $\lambda = 14$

$$\begin{pmatrix} -4 & -2 & -5 \\ -2 & -12 & 3 \\ -5 & 3 & -9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-4X_1 - 2X_2 - 5X_3 = 0$$

$$-2X_1 - 12X_2 + 3X_3 = 0$$

$$-5X_1 + 3X_2 - 9X_3 = 0$$

X_1	X_2	X_3	
-2	-5	4	-2
-12	3	-2	-12
-66	22	44	

$$X_2 = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$$

When $\lambda = 3$

$$\begin{pmatrix} 7 & -2 & -5 \\ -2 & -1 & 3 \\ -5 & 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$7X_1 - 2X_2 - 5X_3 = 0$$

$$-2X_1 - X_2 + 0 = 0$$

$$-5X_1 + 3X_2 + 2X_3 = 0$$

X_1	X_2	X_3	
-2	-5	7	-2
-1	3	-2	-1

$$\begin{array}{ccc} -11 & -11 & -11 \end{array}$$

$$X_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

To find normalized matrix

$$X_1 = \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}$$

$$l = \sqrt{1 + 25 + 16}$$

$$= \sqrt{42}$$

$$N = \begin{pmatrix} 1/\sqrt{42} \\ -5/\sqrt{42} \\ 4/\sqrt{42} \end{pmatrix}$$

$$X_2 = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} \rightarrow N = \begin{pmatrix} -3/\sqrt{14} \\ 1/\sqrt{14} \\ 2/\sqrt{14} \end{pmatrix}$$

$$X_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rightarrow N = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$N = \begin{pmatrix} 1/\sqrt{42} & -3/\sqrt{14} & 1/\sqrt{3} \\ -5/\sqrt{42} & 1/\sqrt{14} & 1/\sqrt{3} \\ 4/\sqrt{42} & 2/\sqrt{14} & 1/\sqrt{3} \end{pmatrix}$$

$$N^T = \begin{pmatrix} 1/\sqrt{42} & -5/\sqrt{42} & 4/\sqrt{42} \\ -3/\sqrt{14} & 1/\sqrt{14} & 2/\sqrt{14} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

$$D = N^T A N$$

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

To find canonical form:

$$(Y_1, Y_2, Y_3) D = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$$

(i.e) ..,

$$(Y_1, Y_2, Y_3) \begin{pmatrix} 0 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}$$

$$14Y_1^2 + 3Y_3^2 = 0$$

Nature:

$$D_1 = 10$$

$$D_2 = \begin{pmatrix} 10 & -2 \\ -2 & 2 \end{pmatrix} = 20 - 4 = 16$$

$$D_3 = 0$$

$\Rightarrow +^{\text{ve}}$ semi definite